Chapter 5 Review

Campbell Biology, AP Edition, 12e

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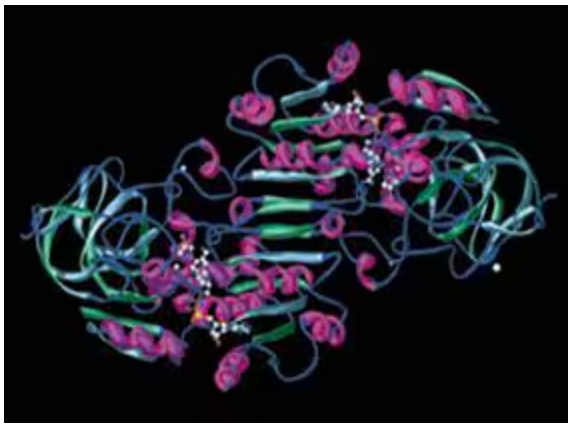


Summary of Key Concepts

How do the sequence and subcomponents of a biological polymer determine its properties? **(BIG IDEA 4)**

Concept 5.1: Macromolecules are polymers, built from monomers

- Large carbohydrates (polysaccharides), proteins, and nucleic acids are **polymers**, chains of **monomers**. Components of lipids vary. Many monomers can form larger molecules by **dehydration reactions**, in which water molecules are released. Polymers can disassemble by the reverse process, **hydrolysis**. An immense variety of polymers can be built from a small set of monomers.

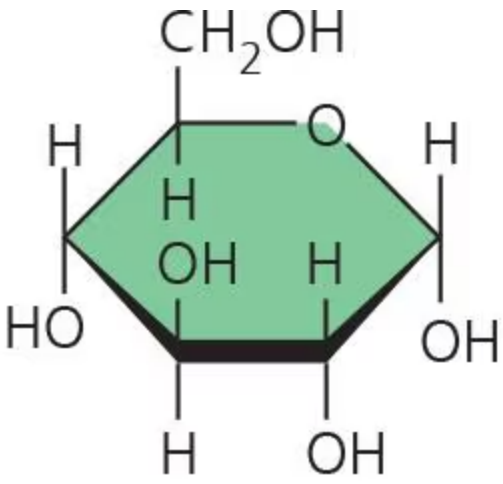
What is the fundamental basis for the differences between large carbohydrates, proteins, and nucleic acids?

Concept 5.2: Carbohydrates serve as fuel and building material

Large Biological Molecules

Compare starch and cellulose. What role does each play in the human body?

Components



Monosaccharide monomer

Abbreviated ring structure. Each unlabeled corner represents a carbon. The ring's thicker edge indicates that you are looking at the ring edge-on; the components attached to the ring lie above or below the plane of the ring. The ring form of glucose is drawn using an edge-on view of a hexagon. The points of the hexagon are on the right and left, and the carbons are numbered starting with the carbon on the far right, which was the aldehyde carbon in the linear form of the sugar. Each carbon is bonded to an O H and a hydrogen, one above the ring, and the other below it. Carbons 2 and 3 are on the near side of the hexagon, carbon 4 is the left point, and carbon 5 is on the far side. The oxygen of carbon 5 completes the hexagon, as it is bonded to

carbon 1. Carbon 6, which is a C H 2 O H is bonded to carbon 5 and drawn above the ring. In glucose, the O H groups are below the ring on carbons 1, 2 and 4, and above the ring on carbon 3. The O H on carbon 1 can be either below or above the ring, depending on whether the ring is in the alpha or beta form, respectively.

Examples

Monosaccharides: glucose, fructose

Disaccharides: lactose, sucrose

Polysaccharides:

- Cellulose (plants)
- Starch (plants)
- Glycogen (animals)
- Chitin (animals and fungi)

Functions

Fuel; carbon sources that can be converted to other molecules or combined into polymers

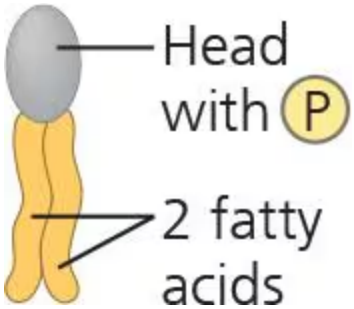
- Cellulose strengthens plant cell walls
- Starch stores glucose for energy in plants
- Glycogen stores glucose for energy in animals
- Chitin strengthens animal exoskeletons and fungal cell walls

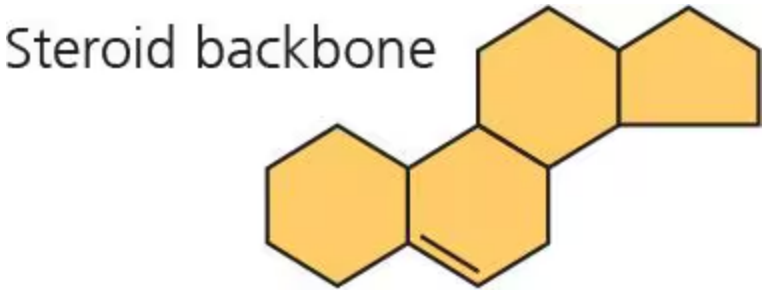
Concept 5.3: Lipids are a diverse group of hydrophobic molecules

Large Biological Molecules

Why are lipids not considered to be polymers or macromolecules?

Components





Examples

*********Triacylglycerols** (fats or oils): glycerol+three fatty acids

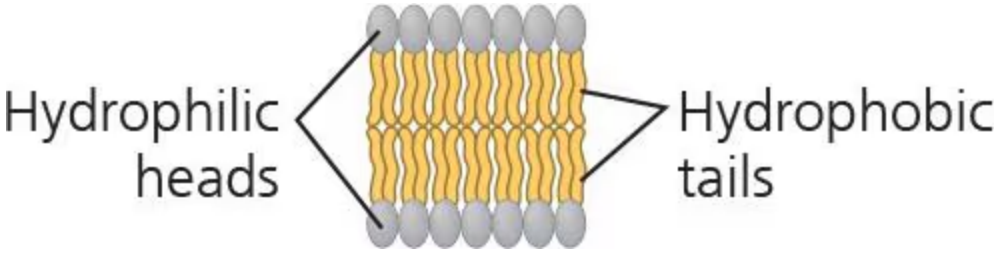
Phospholipids: glycerol+phosphate group+two fatty acids

*********Steroids**: four fused rings with attached chemical groups

Functions

Important energy source

Lipid bilayers of membranes



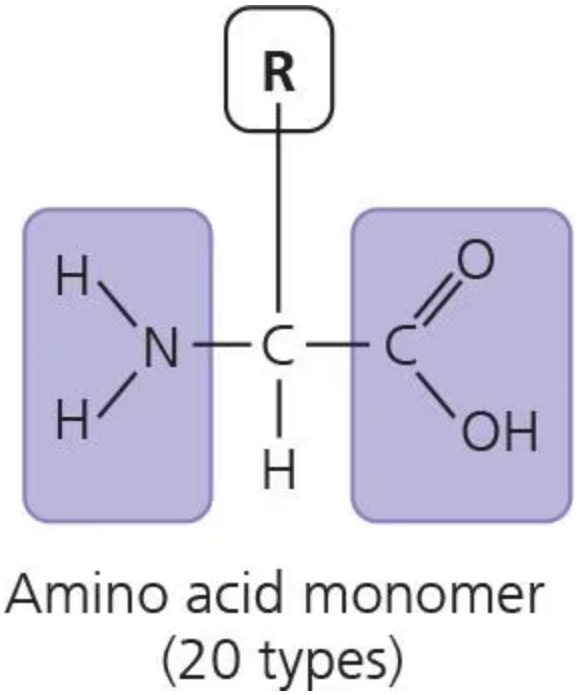
- Component of cell membranes (cholesterol)
- Signaling molecules that travel through the body (hormones)

Concept 5.4: Proteins include a diversity of structures, resulting in a wide range of functions

Large Biological Molecules

Explain the basis for the great diversity of proteins.

Components



Examples

- Enzymes
- Defensive proteins

- Storage proteins
- Transport proteins
- Hormones
- Receptor proteins
- Motor proteins
- Structural proteins

Functions

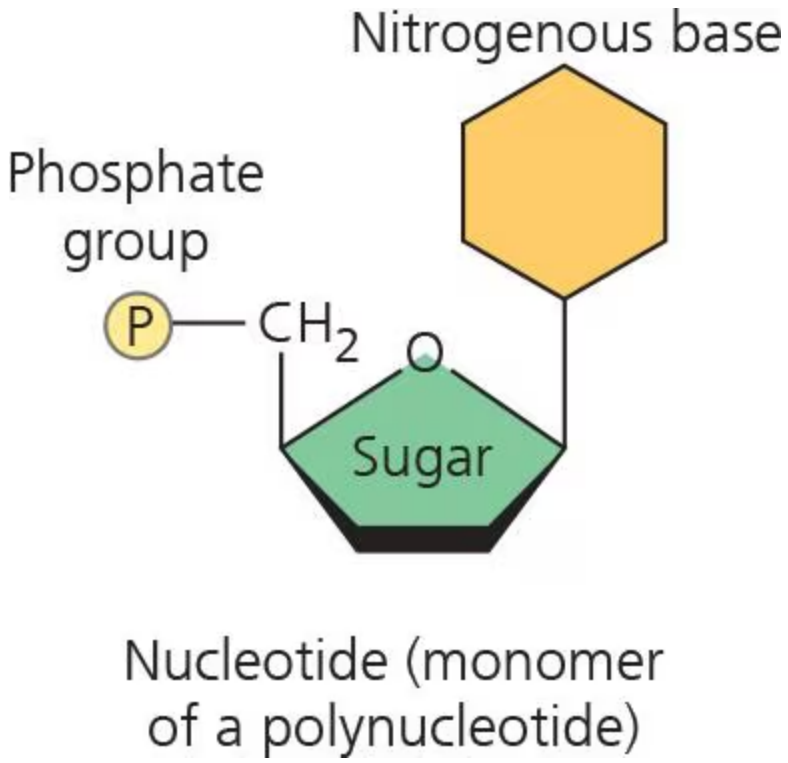
- Catalyze chemical reactions
- Protect against disease
- Store amino acids
- Transport substances
- Coordinate organismal responses
- Receive signals from outside cell
- Function in cell movement
- Provide structural support

Concept 5.5: Nucleic acids store, transmit, and help express hereditary information

Large Biological Molecules

What role does complementary base pairing play in nucleic acids?

Components



Examples

DNA:

- Sugar=deoxyribose
- Nitrogenous bases=C,G,A,T
- Usually double-stranded

RNA:

- Sugar=ribose
- Nitrogenous bases=C,G,A,U
- Usually single-stranded

Functions

Stores hereditary information

Various functions in gene expression, including carrying instructions from DNA to ribosomes

Concept 5.6: Genomics and proteomics have transformed biological inquiry and applications

- Recent technological advances in DNA sequencing have given rise to **genomics**, an approach that analyzes large sets of genes or whole genomes, and **proteomics**, a similar approach for large sets of proteins. **Bioinformatics** is the use of computational tools and computer software to analyze these large data sets.
- The more closely two species are related evolutionarily, the more similar their DNA sequences are. DNA sequence data confirm models of evolution based on fossils and anatomical evidence.

Given the sequences of a particular gene in fruit flies, fish, mice, and humans, predict the relative similarity of the human sequence to that of each of the other species.